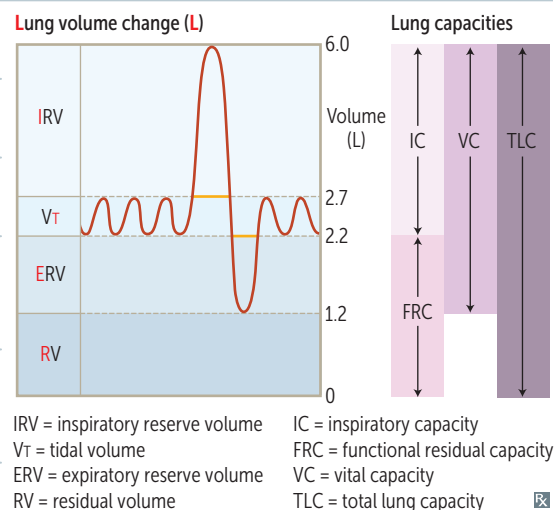


► RESPIRATORY—PHYSIOLOGY

Lung volumes and capacities

There are 4 volumes and 4 capacities. Note: a capacity is a sum of ≥ 2 physiologic volumes.

Tidal volume	Air that moves into lung with each quiet inspiration, 6–8 mL/kg, typically ~500 mL.
Inspiratory reserve volume	Air that can still be breathed in after normal inspiration
Expiratory reserve volume	Air that can still be breathed out after normal expiration
Residual volume	Air in lung after maximal expiration; RV and any lung capacity that includes RV cannot be measured by spirometry
Inspiratory capacity	IRV + V _T Air that can be breathed in after normal exhalation
Functional residual capacity	RV + ERV Volume of gas in lungs after normal expiration; outward pulling force of chest wall is balanced with inward collapsing force of lungs
Vital capacity	IRV + V _T + ERV Maximum volume of gas that can be expired after a maximal inspiration
Total lung capacity	IRV + V _T + ERV + RV = VC + RV Volume of gas present in lungs after a maximal inspiration

**Work of breathing**

Refers to the energy expended or O₂ consumed by respiratory muscles to produce the ventilation needed to meet the body's metabolic demand. Combination of flow-resistive and elastic work (ie, work = force × distance = pressure × volume)—needed to overcome both elastic recoil and airway resistance. Minimized by optimizing respiratory rate (RR) and V_T. ↑ in restrictive diseases (↑ work to overcome elastic recoil resistance achieved with ↑ RR and ↓ V_T) and obstructive diseases (↑ work to overcome airway resistance achieved with ↓ RR and ↑ V_T).

Determination of physiologic dead space

$$V_D = V_T \times \frac{P_{aCO_2} - P_{eCO_2}}{P_{aCO_2}}$$

V_D = physiologic dead space = anatomic dead space of conducting airways plus alveolar dead space; apex of healthy lung is largest contributor of alveolar dead space. V_D = volume of inspired air that does not take part in gas exchange.

P_{aCO₂} = arterial P_{CO₂}.

P_{eCO₂} = expired air P_{CO₂}.

Physiologic dead space—approximately equivalent to anatomic dead space in normal lungs. May be greater than anatomic dead space in lung diseases with ventilation/perfusion mismatch.

Ventilation

Minute ventilation

Abbreviated as V_E . Total volume of gas entering and exiting the lungs per minute.

$$V_E = V_T \times RR$$

Alveolar ventilation

Abbreviated as V_A . Total volume of gas that reaches alveoli each minute.

$$V_A = (V_T - V_D) \times RR$$

Normal values:

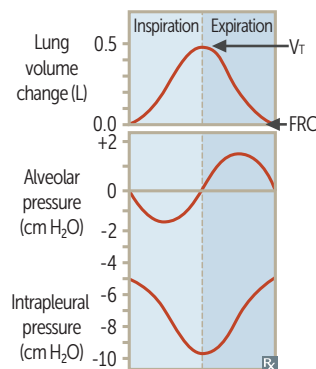
- $RR = 12\text{--}20$ breaths/min
- $V_T = 500$ mL/breath
- $V_D = 150$ mL/breath

Lung and chest wall properties

Lung inflation follows a different pressure-volume curve than lung deflation due to the need to overcome surface tension forces during inflation.

Hysteresis—difference between pressure of inhalation (volume increasing) and pressure of exhalation (volume decreasing).

Because of historical reasons and small pressures, pulmonary pressures are always presented in cm H_2O .



Elastic recoil

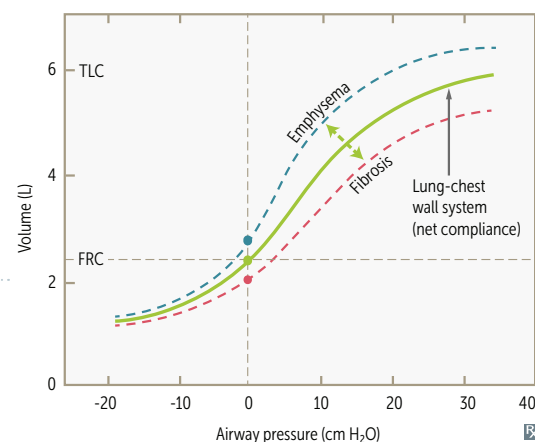
Tendency for lungs to collapse inward and chest wall to spring outward.

At FRC, airway and alveolar pressures equal atmospheric pressure (P_B ; called zero), and intrapleural pressure is negative (preventing atelectasis). The inward pull of the lung is balanced by the outward pull of the chest wall. System pressure is atmospheric. Pulmonary vascular resistance (PVR) is at a minimum.

Compliance

Change in lung volume for a change in pressure ($\Delta V/\Delta P$). Inversely proportional to wall stiffness and increased by surfactant.

- $\uparrow$ compliance = lung easier to fill (eg, emphysema, older adults)
- $\downarrow$ compliance = lung more difficult to fill (eg, pulmonary fibrosis, pneumonia, ARDS, pulmonary edema)



Pulmonary circulation

Normally a low-resistance, high-compliance system. A ↓ in PAO_2 causes hypoxic vasoconstriction that shifts blood away from poorly ventilated regions of lung to well-ventilated regions of lung.

Perfusion limited— O_2 (normal health), CO_2 , N_2O . Gas equilibrates early along the length of the capillary. Exchange can be ↑ only if blood flow ↑.

Diffusion limited— O_2 (emphysema, fibrosis), CO . Gas does not equilibrate by the time blood reaches the end of the capillary.

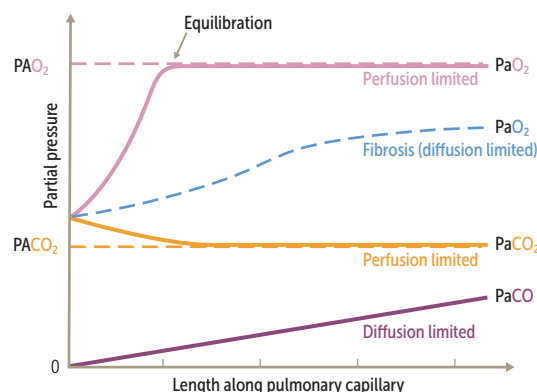
O_2 diffuses slowly, while CO_2 diffuses very rapidly across the alveolar membrane. Disease states that lead to diffusion limitation (eg, pulmonary fibrosis) are more likely to cause early hypoxia than hypercapnia.

Chronic hypoxic vasoconstriction may lead to pulmonary hypertension +/- cor pulmonale.

$$\text{Diffusion } (J) = A \times D_k \times \frac{P_1 - P_2}{\Delta_x} \text{ where}$$

- A = area, Δ_x = alveolar wall thickness,
- D_k = diffusion coefficient of gas,
- $P_1 - P_2$ = difference in partial pressures.
- A ↓ in emphysema.
- Δ_x ↑ in pulmonary fibrosis.

DLCO is the extent to which CO passes from air sacs of lungs into blood.



P_a = partial pressure of gas in pulmonary capillary blood
 P_A = partial pressure of gas in alveolar air

Pulmonary vascular resistance

$$PVR = \frac{P_{\text{pulm artery}} - P_{\text{L atrium}}}{\dot{Q}}$$

Remember: $\Delta P = \dot{Q} \times R$, so $R = \Delta P / \dot{Q}$

$$R = \frac{8\eta l}{\pi r^4}$$

- $P_{\text{pulm artery}}$ = pressure in pulmonary artery
- $P_{\text{L atrium}} \approx$ pulmonary artery occlusion pressure (also called pulmonary capillary wedge pressure)
- $\dot{Q}$ = cardiac output (mL/min)
- R = resistance
- η = viscosity of blood ("stickiness")
- l = vessel length
- r = vessel radius

Ventilation/perfusion mismatch

Ideally, ventilation (V) is matched to perfusion (Q) per minute (ie, $\dot{V}/\dot{Q}$ ratio = 1) for adequate gas exchange.

Lung zones:

- $\dot{V}/\dot{Q}$ at apex of lung = 3 (wasted ventilation)
- $\dot{V}/\dot{Q}$ at base of lung = 0.6 (wasted perfusion)

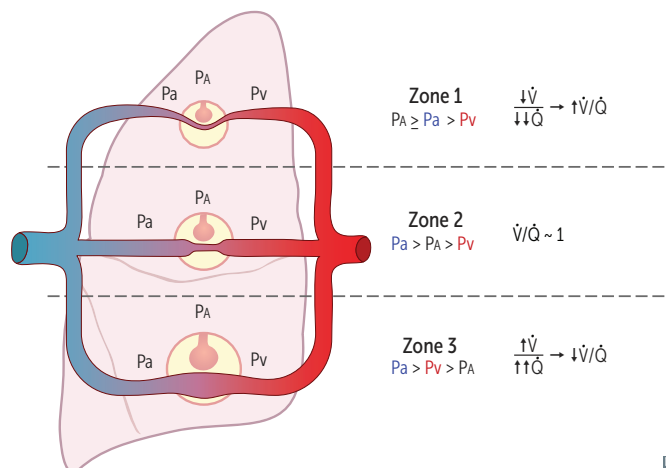
Both ventilation and perfusion are greater at the base of the lung than at the apex of the lung.

With exercise ($\uparrow$ cardiac output), there is vasodilation of apical capillaries $\rightarrow \dot{V}/\dot{Q}$ ratio approaches 1.

Certain organisms that thrive in high O_2 (eg, TB) flourish in the apex.

$\dot{V}/\dot{Q} = 0$ = “airway” obstruction (shunt). In shunt, 100% O_2 does not improve P_{aO_2} (eg, foreign body aspiration).

$\dot{V}/\dot{Q} = \infty$ = blood flow obstruction (physiologic dead space). Assuming $< 100\%$ dead space, 100% O_2 improves P_{aO_2} (eg, pulmonary embolus).

**Alveolar gas equation**

$$P_{AO_2} = P_{IO_2} - \frac{P_{aCO_2}}{RQ}$$

$$\approx 150 \text{ mm Hg}^a - \frac{P_{aCO_2}}{0.8}$$

^aAt sea level breathing room air

P_{AO_2} = alveolar PO_2 (mm Hg)

P_{IO_2} = PO_2 in inspired air (mm Hg)

P_{aCO_2} = arterial PCO_2 (mm Hg)

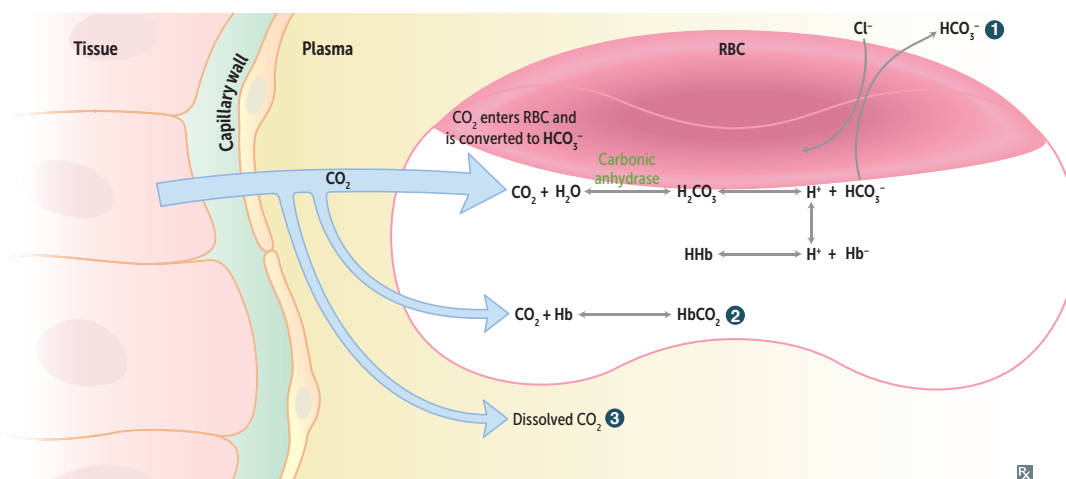
RQ = respiratory quotient = CO_2 produced/
 O_2 consumed

A-a gradient = $P_{AO_2} - P_{aO_2}$. Normal A-a gradient estimated as $(\text{age}/4) + 4$ (eg, for a person < 40 years old, gradient should be < 14).

Carbon dioxide transport

Majority of CO_2 (90%) must be converted to HCO_3^- in RBCs ❶ → carried in the plasma to the lungs (and released via the carbonic anhydrase reaction). CO_2 (5%) also binds to various plasma proteins (carbamino compounds) including the N-terminus of globin in deoxygenated hemoglobin in RBCs → carbaminohemoglobin or HbCO_2 ❷. Small percentage of CO_2 (5%) can be dissolved into plasma itself ❸.

In the lungs, Hb oxygenation promotes dissociation of H^+ → equilibrium shifts towards CO_2 production → CO_2 is released from RBCs (Haldane effect).



Hypoxia and hypoxemia

Hypoxia

↓ O_2 delivery to tissues. Commonly due to ↓ cardiac output, hypoxemia (insufficient oxygenation of blood with ↓ PaO_2), ischemia, anemia, CO/ cyanide poisoning.

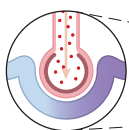
❶ ↓ inspired oxygen tension (P_{IO_2})

$\text{P}_{\text{IO}_2} = \text{F}_{\text{IO}_2} \times (\text{P}_{\text{B}} - \text{P}_{\text{H}_2\text{O}})$: most commonly due to ↓ P_{B} in high altitude

❷ Hypoventilation (due to ↑ Paco_2)

$\text{PAO}_2 = \text{P}_{\text{IO}_2} - \text{Paco}_2 / \text{RQ}$ (eg, CNS depression from opiate overdose, obesity hypoventilation syndrome, neuromuscular weakness)

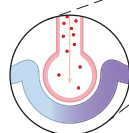
❸ Diffusion limitation (eg, fibrosis)



Normal A-a gradient
Increased A-a gradient

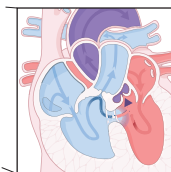
❹ $\dot{V}/\dot{Q}$ mismatch

Normal perfusion (edema), or ↓ perfusion in areas of normal ventilation (eg, pulmonary embolism)



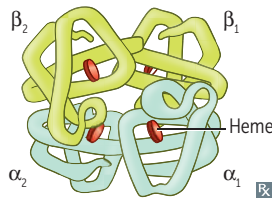
❺ Right-to-left shunt (the extreme of $\dot{V}/\dot{Q}$ mismatch)

Normal perfusion in areas of no ventilation. Can be anatomic (eg, intracardiac shunt) or physiologic (eg, perfusion of nonventilated alveoli in ARDS)



Hypoxemia

Insufficient oxygenation of blood (↓ PaO_2).

Hemoglobin

Normal adult hemoglobin (Hb) is composed of 4 polypeptide subunits (2 α and 2 β) that each bind one O_2 molecule. Hb is an allosteric protein that exhibits positive cooperativity when binding to O_2 , such that:

- Oxygenated Hb has high affinity for O_2 (300 $\times$).
- Deoxygenated Hb has low affinity for $O_2 \rightarrow$ promotes release/unloading of O_2 .

The protein component of hemoglobin acts as buffer for H^+ ions and CO_2 .

Myoglobin is composed of a single polypeptide chain associated with one heme moiety. Higher affinity for oxygen than Hb. Essential for full oxygenation of aerobically active muscle.

Oxygen content of blood

$$O_2 \text{ content} = (O_2 \text{ bound to hemoglobin}) + (O_2 \text{ solubilized in plasma}) = (1.34 \times Hb \times SaO_2) + (0.003 \times PaO_2).$$

SaO_2 = percent saturation of arterial blood with O_2 .

0.003 = solubility constant of O_2 ; PaO_2 = partial pressure of O_2 in arterial blood.

Normally 1 g Hb can bind 1.34 mL O_2 ; normal Hb amount in blood is 15 g/dL.

O_2 binding (carrying) capacity $\approx$ 20 mL O_2 /dL of blood.

With $\downarrow$ Hb there is $\downarrow$ O_2 content of arterial blood, but no change in O_2 saturation and PaO_2 .

O_2 delivery to tissues = cardiac output $\times$ O_2 content of blood.

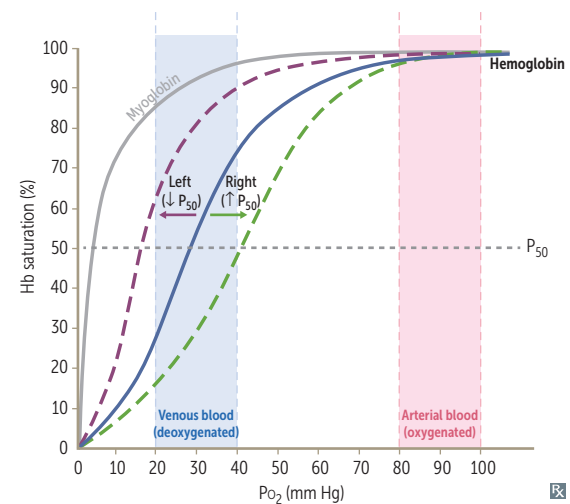
	Hb CONCENTRATION	SaO_2	PaO_2	TOTAL O_2 CONTENT
CO poisoning	Normal	$\downarrow$ (CO competes with O_2)	Normal	$\downarrow$
Anemia	$\downarrow$	Normal	Normal	$\downarrow$
Polycythemia	$\uparrow$	Normal	Normal	$\uparrow$
Methemoglobinemia	Normal	$\downarrow$ (Fe^{3+} poor at binding O_2)	Normal	$\downarrow$
Cyanide toxicity	Normal	Normal	Normal	Normal

Oxyhemoglobin dissociation curve

Shifts in oxyhemoglobin dissociation curve (ODC) reflect local tissue oxygen needs. Can be helpful (meets metabolic needs) or harmful (in toxicities, pathophysiologic situations).

Right shift in ODC reflects $\downarrow$ Hb affinity for $O_2 \rightarrow \uparrow$ O_2 unloading at tissue. Physiologically occurs with $\uparrow$ O_2 needs: exercise, $\downarrow$ pH, $\uparrow$ temperature/fever, hypoxia ($\uparrow$ 2,3-BPG); at the cellular level, caused by $\uparrow$ H^+ and $\uparrow$ CO_2 created by tissue metabolism (Bohr effect).

Left shift in ODC reflects $\uparrow$ Hb affinity for $O_2 \rightarrow \downarrow$ O_2 unloading at tissue. Physiologically occurs with $\downarrow$ O_2 needs ($\downarrow$ temperature) and pregnancy (fetal Hb has higher O_2 affinity than adult Hb, and $\uparrow$ O_2 binding due to $\downarrow$ affinity for 2,3-BPG $\rightarrow$ left shift, driving O_2 across placenta to fetus). Pathologically occurs with $\uparrow$ CO, $\uparrow$ MetHb, genetic mutation ($\downarrow$ 2,3-BPG). **Left is lower.**



ODC has sigmoidal shape due to positive cooperativity (ie, tetrameric Hb molecule can bind 4 O_2 molecules and has higher affinity for each subsequent O_2 molecule bound). Myoglobin is monomeric and thus does not show positive cooperativity; curve lacks sigmoidal appearance.

Response to high altitude

Constant F_{IO_2} but $\downarrow P_B \rightarrow \downarrow$ atmospheric oxygen (P_{IO_2}) $\rightarrow \downarrow P_{aO_2} \rightarrow \uparrow$ ventilation $\rightarrow \downarrow P_{aCO_2}$ $\rightarrow$ respiratory alkalosis $\rightarrow$ altitude sickness (headaches, nausea, fatigue, lightheadedness, sleep disturbance).

Chronic $\uparrow$ in ventilation.

$\uparrow$ erythropoietin (primarily from kidneys) $\rightarrow \uparrow$ Hct and Hb (due to chronic hypoxemia).

$\uparrow$ 2,3-BPG (binds to Hb $\rightarrow$ rightward shift of oxyhemoglobin dissociation curve $\rightarrow \uparrow O_2$ release) due to increased glycolysis and isomerization by BPG mutase.

Cellular changes ($\uparrow$ mitochondria).

$\uparrow$ renal excretion of HCO_3^- to compensate for respiratory alkalosis (can augment with acetazolamide).

Chronic hypoxic pulmonary vasoconstriction $\rightarrow \uparrow$ pulmonary vascular resistance $\rightarrow$ pulmonary hypertension, right ventricular hypertrophy (RVH).

Response to exercise

$\uparrow$ HR and $\uparrow$ SV $\rightarrow \uparrow \dot{Q} \rightarrow \uparrow$ pulmonary blood flow $\rightarrow \uparrow \dot{V}/\dot{Q}$ ratio from base to apex (becoming more uniform).

$\uparrow$ cellular respiration $\rightarrow \uparrow CO_2$ production and $\downarrow$ pH at tissues $\rightarrow$ right shift of ODC $\rightarrow$ tissue offloading of more $O_2 \rightarrow \uparrow O_2$ consumption ($\uparrow O_2$ difference in arteries and veins). $\uparrow$ RR to meet $\uparrow O_2$ demand and remove excess $CO_2 \rightarrow \uparrow$ pulmonary blood flow.

P_{aO_2} and P_{aCO_2} are maintained by homeostatic mechanisms.

$\downarrow P\bar{v}O_2$ due to $\uparrow O_2$ consumption.

$\uparrow P\bar{v}CO_2$ due to $\uparrow CO_2$ production.

Methemoglobin

Iron in Hb is normally in a reduced state (ferrous Fe^{2+} ; “just the **2** of **us**”). Oxidized form of Hb (ferric, Fe^{3+}) has reduced O_2 affinity $\rightarrow$ tissue hypoxia from $\downarrow O_2$ saturation and $\downarrow O_2$ content.

Fe^{3+} also has $\uparrow$ affinity for cyanide. This oxidized form is called methemoglobinemia.

While typical concentrations are 1–2%, methemoglobinemia will occur at higher levels and may present with cyanosis (does not improve with supplemental O_2) and with chocolate-colored blood.

Dapsone, local anesthetics (eg, benzocaine), and nitrites (eg, from dietary intake or polluted water sources) cause poisoning by oxidizing Fe^{2+} to Fe^{3+} .

Methemoglobinemia can be treated with **methylen**e blue and vitamin C.

Cyanide vs carbon monoxide poisoning

Both inhibit aerobic metabolism via inhibition of complex IV of ETC (cytochrome c oxidase) → hypoxia that does not fully correct with supplemental O₂ and ↑ anaerobic metabolism.

	Cyanide	Carbon monoxide
EXPOSURE	Synthetic product combustion, amygdalin ingestion (found in apricot seeds), cyanide ingestion (eg, in suicide attempts), fire victims. Risk of cyanide toxicity with use of nitroprusside in hypertensive emergencies.	From tobacco smoke, furnaces, space heaters, fires, motor exhaust (incomplete combustion of carbon-containing compounds). Odorless, tasteless, colorless, non-irritating. Leading worldwide cause of death by poisoning.
PRESENTATION	Headache, dyspnea, drowsiness, seizure, coma. Skin may appear flushed (“cherry red”) due to bright red venous blood. Venules in retina appear bright red. Breath may have bitter almond odor.	Headache, vomiting, confusion, visual disturbances, coma. May have cherry-red skin with bullous skin lesions. Multiple victims may be involved (eg, family due to faulty furnace).
LABS	Normal PaO ₂ . ↑ lactate → anion gap metabolic acidosis.	Normal PaO ₂ . ↑ carboxyhemoglobin on co-oximetry (cannot be distinguished with pulse oximetry). Classically associated with bilateral globus pallidus lesions on MRI A , although can rarely be seen with cyanide toxicity.
EFFECT ON OXYGEN-HEMOGLOBIN CURVE	Cyanide binds cytochrome a3 in complex IV → ample O ₂ but cannot be used due to ineffective oxidative phosphorylation. Curve normal. O ₂ saturation may appear normal initially.	Left shift in ODC → ↑ affinity for O ₂ → ↓ O ₂ unloading in tissues. Binds competitively to Hb with > 200× greater affinity than O ₂ to form carboxyhemoglobin → ↓ %O ₂ saturation of Hb.
TREATMENT	Decontamination (eg, remove clothing). 100% O ₂ is ineffective; instead give treatments to remove and excrete the cyanide: - Hydroxocobalamin (binds cyanide → cyanocobalamin → renal excretion) - Nitrites (oxidize Hb → methemoglobin → binds cyanide → cyanomethemoglobin → ↓ toxicity) - Sodium thiosulfate (↑ cyanide conversion to thiocyanate → renal excretion)	Give 100% O ₂ to overcome the increased affinity for CO. Hyperbaric oxygen if severe. CO-Hb half-life is ~300 mins → ↓ to ~80 mins on 100% O ₂ → ↓ to ~20 mins in a hyperbaric O ₂ chamber. If concurrent CO and cyanide poisoning are suspected (eg, in victims of a fire), give hydroxocobalamin, rather than nitrites or sodium thiosulfate, to avoid increasing methylglobin.

